VENDOR APPROVAL & PRODUCT SUBMISSION

PulmOne MiniBox+ — Patchogue Pulmonary Clinic

Date: May 12, 2026

Prepared for: Kennedy Wright, Stony Brook Medicine

Intended audience: Stony Brook Medicine corporate procurement, vendor management, and IT review teams

Submitted by: Erik Bernard, NYC Metro Sales Specialist, PulmOne USA Inc.

Purpose of This Submission

Per the request of Kennedy Wright at Stony Brook Medicine, this package contains the technical, clinical, regulatory, and corporate documentation needed for Stony Brook Medicine's corporate office to evaluate PulmOne USA Inc. as a new approved vendor and the PulmOne MiniBox+ as a clinically approved pulmonary function testing (PFT) device for use at the Patchogue pulmonary clinic.

Everything Stony Brook's corporate procurement, vendor management, and IT review teams will typically require to onboard PulmOne and approve the MiniBox+ for capital purchase is enclosed in the seven (7) tabs that follow.

About the PulmOne MiniBox+

The PulmOne MiniBox+ is a desktop, gasless, cabinless plethysmography system that performs complete pulmonary function testing — spirometry, lung volumes (TLC, RV, FRC), and single-breath CO diffusion (DLCO) — in a single 18-pound device. It is FDA-cleared, IEC 60601-1 compliant, and fully ATS/ERS guideline compliant. Lung volume measurements have been independently validated as equivalent to traditional body box plethysmography in a multi-center peer-reviewed study published in CHEST Journal (Berger et al., February 2021), with normalized standard deviation of just 7.0% in healthy subjects.

MiniBox+ requires no specialized testing booth, accommodates obese and wheelchair-bound patients, completes a full PFT in approximately 20 minutes, and integrates with all major EHR systems (Epic, Cerner, and others) via HL7. The sole-source justification letter (Tab 4) details why this product is uniquely qualified for the Patchogue practice's clinical and operational requirements.

Quote

A formal capital equipment quote will be provided separately through PulmOne's standard ordering process. This submission contains the technical, clinical, and corporate documentation only.

Point of Contact

For any technical, clinical, regulatory, or contractual questions, please contact the undersigned directly. I can also arrange a call or web demonstration with PulmOne's clinical, regulatory, or IT integration teams at any time.

PACKAGE INDEX

Documents Enclosed

The following seven documents accompany this cover letter. They have been compiled to address every standard requirement of an IDN corporate procurement, vendor management, and IT review process.

Tab	Document	Purpose for Corporate Review
1	**Product Brochure & Technical Specifications**	Overview of the MiniBox+ platform, full technical specifications, IEC 60601-1 compliance, dimensions, power, and reimbursable CPT codes.
2	**Competitive Comparison Matrix**	Side-by-side comparison of MiniBox+ vs. traditional body box plethysmography and vs. nitrogen washout. Demonstrates clinical and operational advantages.
3	**Clinical Validation — CHEST Journal (Peer-Reviewed)**	Multi-center validation study (Berger et al., CHEST Journal, Feb 2021) demonstrating MiniBox+ lung volume measurements are equivalent to body box plethysmography. n=266 across 5 medical centers in the US and Europe.
4	**Sole Source Justification Letter**	Manufacturer's sole-source letter establishing PulmOne MiniBox+ as the only desktop, gasless, cabinless plethysmography system capable of full PFT. Supports a no-bid capital purchase decision.
5	**HL7 Conformance & Cybersecurity Documentation**	Full HL7 v2.5.1 conformance statement for EHR/EMR integration (Epic, Cerner, and other major systems). Includes security architecture: BitLocker encryption, password controls, account lockout, and auto-logout. For Stony Brook IT review.
6	**PulmOne USA Inc. — IRS Form W-9**	Manufacturer's signed W-9 for vendor master file setup. PulmOne USA Inc., EIN 47-4165945, 444 Madison Avenue, New York, NY 10022. C-Corporation.
7	**User Manual (Rev. 20) — Provided Separately**	Complete operator's manual (185 pages). Included in the digital package as a separate file due to size. Available on request for biomedical engineering review.

Respectfully submitted,

Erik Bernard
NYC Metro Sales Specialist

PulmOne USA Inc.
Erik.Bernard@pulm-one.com
444 Madison Avenue, 6th Floor, New York, NY 10022

PulmOne
PFT Outside The Box

MiniBox+™

Complete Pulmonary Function Testing.
Because Spirometry is just not enough.

EASY

Easy to operate with minimum set-up. Fully automatic Lung Volumes and DLCO measurements. All results automatically generated.

ACCURATE

Clinically validated in over 500 patients at 8 medical centers, based on comparative data from leading body plethysmography devices.

GREAT ROI

With all measurements fully reimbursable, you can achieve your ROI within months, with testing of only 2 to 5 patients daily.

Also available in MiniBox® configuration for spirometry and lung volumes only

MiniBox+

Complete, All-in-One Pulmonary Function Testing (PFT)

Desktop Plethysmography, Spirometry & Diffusion Testing

Easy to use, MiniBox+ provides accurate and repeatable, fully automatic measurements of Total Lung Capacity (TLC) and Residual Volume (RV), as well as Spirometry and Single Breath CO Diffusion (DLCO).

FULLY COMPLIANT WITH ATS/ERS GUIDELINES:

SPIROMETRY

Complete set of 30+ parameters including FEV1, FVC, SVC, MVV

LUNG VOLUMES

Same volumes as body plethysmography including TLC, FRC, RV

DIFFUSING CAPACITY

DLCO, VA, DLCO/VA
Single breath method

* Reference values may not be available for DLCO for children under the age of 7

OTHER MEASUREMENTS

6MWT, Provocation

INNOVATIVE

Fully-automated Quality Control algorithms enable ease-of use and reliability

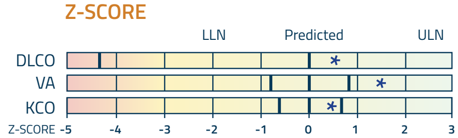
FULLY AUTOMATIC LVM AND DLCO

Manual measurement optional

ACCURATE & REPEATABLE

Automatically generated results

Z-SCORE



USER-FRIENDLY

Intuitive user interface, customizable reports, and a wide range of data export options

FULL SELECTION OF PREDICTED EQUATIONS

GLI reference values are standard for both Spirometry and Diffusion but easily customizable

PORTABLE

Lightweight with small footprint; Optional cart

PEDIATRIC USE

Measurements can be taken on children from age 5

GASLESS & CABINLESS PLETHYSMOGRAPHY

No sensation of claustrophobia; Can test obese, wheelchair-bound, or non-ambulatory patients

SELF-CALIBRATING

Due to unique sensor technology

CONNECTIVITY (OPTIONAL)

HL7 compliant, easily configurable, integrates with all major EHR systems

FeNO by NIOX®

ADD-ON (OPTIONAL)

Fully integrated

The NIOX VERO device is attached to the MiniBox+ for transportation purposes only. FeNO tests should not be performed while attached to the MiniBox+.

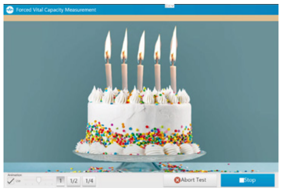
With as little as 60 seconds of tidal breathing required for Lung Volume Measurements, this gasless, cabinless system provides maximum patient comfort.

EASE OF USE

- Automatic zeroing of analyzers before each measurement
- No stabilization time for plethysmography
- Warm-up time for analyzer less than 10 min
- Volume Gas graph for diffusion measurement
- Fast analyzer with response time t0-90 in less than 85 ms
- Time-to-test: as little as 20 minutes with pre- and post-spirometry

FLOW VOLUME

INCENTIVES



Complete mobility with dedicated MiniBox+ cart

MiniBox+ Technical Specifications

General

Volume accuracy	±2%
Flow accuracy	±2.5% or 50 mL/s
Flow range	±14 L/s
Pressure range	±1 PSI, 7 kPa, Bi-directional
Dynamic resistance at 12L/s (spirometer)	<1.2 cmH2O/(L/s)

Standards Compliance

Electrical	IEC 60601-1
EMC	IEC 60601-1-2

Physical

Full HD screen	10.8"
Dimensions	510x470x310 mm/ 20x18.5x12 in
Weight	8 kg / 18 lbs

Valve

Closing duration	100 ms

Electrical

Power	100-240VAC,50-60Hz

Gas Analyzer

Spectrometer	Infrared (CO,CH4)
Range	0-0.35%
Calibration	Automatic with quality control
Response time	t0-90 85 ms

For a complete list of technical specs

MiniBox+ changes the economics of your practice

With MiniBox+, you can expand your medical practice while making Pulmonary Function Testing (PFT) more accessible to patients. Reimbursement for all testing measurements ensures a viable model for better patient care and better economics.

MKT008 rev.08

No. measurements/day	Potential reimbursement/month *	Potential ROI
	$17,000 5 Patients Per Day	1.7 Months
	$13,600 4 Patients Per Day	2.1 Months
	$10,200 3 Patients Per Day	2.9 Months
	$6,800 2 Patients Per Day	4.3 Months
	$3,400 1 Patient Per Day	8.7 Months

FLEXIBLE FINANCING PLANS ARE AVAILABLE

Ranging from $340 to $570 monthly payments over 3 to 5 years, with an option of starting payment 3 months post-purchase.

Consult with your financial advisor about the Section 179 tax benefits associated with the purchase of MiniBox+.

All tests are eligible for reimbursement*:

*subject to specific health plan policy

Code	Test
94010	Forced vital capacity - breathing capacity test
94060	Pre vs. Post bronchodilator - breathing capacity test
94200	Maximum breathing capacity
94726	Total lung capacity - determination of lung volume/plethysmography
94729	Diffusing Capacity
94375	Respiratory flow-volume loop
94618	Pulmonary stress testing (6-Minute-Walk-Test)
95012	Fractional exhaled nitric oxide (Add-on)

Coding should be based on medical necessity and is not intended to maximize reimbursement.

Expand your practice today. With MiniBox+.

ABOUT US

PulmOne is a global medical technology company dedicated to providing innovative solutions for complete Pulmonary Function Testing (PFT) in any clinical setting. At PulmOne, we develop patient-friendly, hassle-free, and budget-conscious devices that deliver a wide range of accurate measurements for the diagnosis, treatment, and monitoring of respiratory diseases.

PulmOne USA Tel: 1-844-PulmOne (1-844-785-6663)
Email us: usinfo@pulm-one.com
Visit us: www.pulm-one.com
PulmOne Advanced Medical Devices, Ltd. 14 Hamelacha St., Ra'anana 4366107 Israel

PulmOne
PFT Outside The Box

How does MiniBox+™ Plethysmography compare to Nitrogen Washout?

	Nitrogen Washout	MiniBox+
Complete PFT testing time	Up to 60 min.	As little as 20 min.
Easy set-up and operation	✕	✓
Low Maintenance	✕	✓
Measures Total Lung Volume	✕	✓
Lung Volume Measurement: Fully automatic	✕	✓
Lung Volume Measurement: Gasless	✕	✓
Lung Volume Measurement Testing Time	Up to 30 min.	As little as 60 sec.
Not impacted by draft	✕	✓
No wait-time between efforts	✕	✓

PulmOne USA Tel: 1-844-PulmOne (1-844-785-6663) **Email us:** usinfo@pulm-one.com **Visit us:** www.pulm-one.com
PulmOne Advanced Medical Devices, Ltd. 14 Hamelacha St., Ra'anana 4366107 Israel

PulmOne
PFT Outside The Box

How does MiniBox+ Plethysmography compare to Body Box Plethysmography?

	Body Box Plethysmography	MiniBox+ Plethysmography
Performs Spirometry, Lung Volume Plethysmography & DLCO	✓	✓
Complete PFT testing time	30-45 min.	As little as 20 min.
Easy set-up and operation	✕	✓
Desktop or cart-mounted	✕	✓
Portable – can be moved between rooms	✕	✓
Lung Volume Measurement: Fully automatic	✕	✓
Lung Volume Measurement: As little as 60 sec.	✕	✓
Lung Volume Measurement: Relaxed Tidal Breathing	✕	✓
No wait-time between efforts	✕	✓
ROI within 7 weeks (at 5 patients per day)*	✕	✓
Can easily test wheelchair-bound patients	✕	✓
Can easily test obese patients	✕	✓
Can be easily moved for bedside testing	✕	✓

*subject to specific health plan policy

PUO-DOC20027 REV.02

PulmOne USA Tel: 1-844-PulmOne (1-844-785-6663) **Email us:** usinfo@pulm-one.com **Visit us:** www.pulm-one.com
PulmOne Advanced Medical Devices, Ltd. 14 Hamelacha St., Ra'anana 4366107 Israel

PulmOne
PFT Outside The Box

CHEST

Validation of a Novel Compact System for the Measurement of Lung Volumes

Kenneth I. Berger, MD; Ori Adam, PhD; Roberto Walter Dal Negro, MD; David A. Kaminsky, MD; Robert J. Shiner, MD; Felip Burgos, PhD; Frans H. C. de Jongh, PhD; Inon Cohen, PhD; and Jeffrey J. Fredberg, PhD

BACKGROUND: Current techniques for measuring absolute lung volumes rely on bulky and expensive equipment and are complicated to use for the operator and the patient. A novel method for measurement of absolute lung volumes, the MiniBox method, is presented.

RESEARCH QUESTION: Across a population of patients and healthy participants, do values for total lung capacity (TLC) determined by the novel compact device (MiniBox, PulmOne Advanced Medical Devices, Ltd.) compare favorably with measurements determined by traditional whole body plethysmography?

STUDY DESIGN AND METHODS: A total of 266 participants (130 men) and respiratory patients were recruited from five global centers (three in Europe and two in the United States). The study population comprised individuals with obstructive (n = 197) and restrictive (n = 33) disorders as well as healthy participants (n = 36). TLC measured by conventional plethysmography (TLC_{Pleth}) was compared with TLC measured by the MiniBox (TLC_{MB}).

RESULTS: TLC values ranged between 2.7 and 10.9 L. The normalized root mean square difference (NSD) between TLC_{Pleth} and TLC_{MB} was 7.0% in healthy participants. In obstructed patients, the NSD was 7.9% in mild obstruction and 9.1% in severe obstruction. In restricted patients, the NSD was 7.8% in mild restriction and 13.9% in moderate and severe restriction. No significant differences were found between TLC values obtained by the two measurement techniques. Also no significant differences were found in results obtained among the five centers.

INTERPRETATION: TLC as measured by the novel MiniBox system is not significantly different from TLC measured by conventional whole body plethysmography, thus validating the MiniBox method as a reliable method to measure absolute lung volumes.

CHEST 2021; 159(6):2356-2365

KEY WORDS: novel measurement; physiology; plethysmograph; pulmonary function test; total lung capacity

FOR EDITORIAL COMMENT, SEE PAGE 2143

ABBREVIATIONS: ATS = American Thoracic Society; ERS = European Respiratory Society; FRC = functional residual capacity; NSD = normalized root mean square difference; RV = residual volume; TLC = total lung capacity; TLC_{MB} = total lung capacity measured by the MinBox; TLC_{Pleth} = total lung capacity measured by plethysmography

AFFILIATIONS: From the Division of Pulmonary Critical Care and Sleep Medicine (K. I. Berger), NYU Grossman School of Medicine, the André Cournand Pulmonary Physiology Laboratory (K. I. Berger), Bellevue Hospital, New York, NY; the Institute of Earth Sciences (O. Adam), Hebrew University, Jerusalem, Israel; the Centro Nazionale Studi di Farmacoeconomia e Farmacoepidemiologia Respiratoria (R. W. Dal Negro), CESFAR, Verona, Italy; Pulmonary and Critical Care Medicine (D. A. Kaminsky), The University of Vermont Larner College of Medicine, Burlington, VT; the Herzliya Medical Center (R. J. Shiner), Herzliya Pituach, Israel; Servicio de Pneumologia (F. Burgos), Hospital Clínic, Instituto de Investigaciones Biomédicas August Pi i Sunyer (IDIBAPS), University of Barcelona, Barcelona, Spain; the Department of Pulmonary Function (F. H. C. de Jongh), Medisch Spectrum Twente, Enschede, The Netherlands; no institutional affiliation (I. Cohen); and the Department of Environmental Health (J. J. Fredberg), Harvard T.H. Chan School of Public Health, Boston, MA.

FUNDING/SUPPORT: The study was sponsored by PulmOne Advanced Medical Devices, Ltd., at the Burlington, Vermont; Barcelona, Spain; Enschede, The Netherlands; and New York, New York, sites.

CORRESPONDENCE TO: Kenneth I. Berger, MD; e-mail: Kenneth.berger@nyumc.org

DOI: https://doi.org/10.1016/j.chest.2021.01.052

Take-home Points

Study Question: Across a population of patients and healthy participants, do values for total lung capacity (TLC) determined during tidal breathing by a novel compact device (MiniBox) compare favorably with measurements determined by traditional whole body plethysmography?
Results: The novel MiniBox system yielded accurate and reproducible determination of TLC as compared with the current gold standard plethysmography technique (normalized standard difference, 8.5%; mean discrepancy, –0.05 L) without the need for ionizing radiation or inhalation of inert gases.
Interpretation: The MiniBox system should be considered a clinically reliable technique to obtain measurement of lung volumes that are equivalent to plethysmography in both healthy participants as well as those with lung disease.

Measurement of absolute lung volumes, specifically residual volume (RV), functional residual capacity (FRC), and total lung capacity (TLC), are useful in the diagnosis and management of respiratory disorders as a supplement to simple spirometry. However, determination of these volumes requires more complex and expensive technologies than those used for spirometry. Consequently, many clinics in private and hospital settings rely on results of spirometry alone when assessing lung function, which is not in accord with American Thoracic Society (ATS)/European Respiratory Society (ERS) guidelines.[1]

Five methods currently are available for measuring absolute lung volume: whole body plethysmography, multibreath helium dilution, nitrogen washout, CT scanning, and chest radiography. Each technique has advantages and disadvantages to the clinician and the patient. CT scanning and chest radiography are not used in pulmonary function laboratories and incur radiation exposure, whereas helium dilution and nitrogen washout may underestimate lung volumes because gas may not distribute fully to poorly ventilated areas. Body plethysmography may overestimate lung volume relative to other measurements,[2] primarily in the setting of airflow obstruction and increased compliance of the extrathoracic airway.[3] Several comparative studies have demonstrated differences between the gas-based techniques (helium dilution and nitrogen washout) and plethysmography with a normalized root mean square difference (NSD) ranging between 8.8% and 23.7% (Fig 1).[2-11]

Whole body plethysmography is simple in principle but inherently complex in practice because patients must sit inside a sealed booth and perform a complex respiratory maneuver against a closed shutter (ie, an occluded airway). Although gas dilution and gas washout techniques are well-established alternatives to plethysmography, they require more time to demonstrate repeatability between maneuvers, especially in patients with obstructive airway diseases. Moreover, these gas-based techniques correlate well with plethysmography only in healthy participants, but underestimate lung volumes in patients with airflow obstruction.[2,3,6,7,12]

Until recently, the search for practical alternatives to plethysmography has remained unsuccessful. Imaging techniques are not practical everyday tools because of availability, cost, and radiation exposure.[2,8-10] Respiratory system impedance, even when extended to a wide range of forcing frequencies, cannot yield estimates for absolute lung volumes.[13-15] Similarly, forced expiratory maneuver techniques are inadequate.[5] These failures may reflect the complex dynamics of gas distribution within the lung, particularly in participants with obstructive diseases.

The MiniBox (PulmOne Advanced Medical Devices, Ltd.) is a new Food and Drug Administration-approved device that is not yet included in ATS/ERS guidelines.[16] The MiniBox is a table-top unit that includes a flow-interruption device and a rigid container (Fig 2). Like plethysmography, it measures lung volumes in a manner that does not require radiation, forced maneuvers, or inhalation of inert gases. The MiniBox derives TLC during tidal breathing by analysis of gas pressures and air flows immediately preceding and following airway occlusions based on a combination of first principles and inductive statistics. The present study was designed to evaluate the measurement bias and equivalency between TLC determined by the novel compact device (MiniBox) and measurements determined by traditional whole body plethysmography across a population of patients and healthy participants.

Methods

To validate the MiniBox technique we recruited an international group of investigators based in Europe (Enschede, The Netherlands; Barcelona, Spain) and North America (New York, New York, and Burlington, Vermont) and compared TLC measured by the MiniBox (TLC_{MB}) with TLC measured by plethysmography (TLC_{Pleth}). A nonsponsored center (Verona, Italy) also participated in the study and adhered to the same protocol as the other four centers.

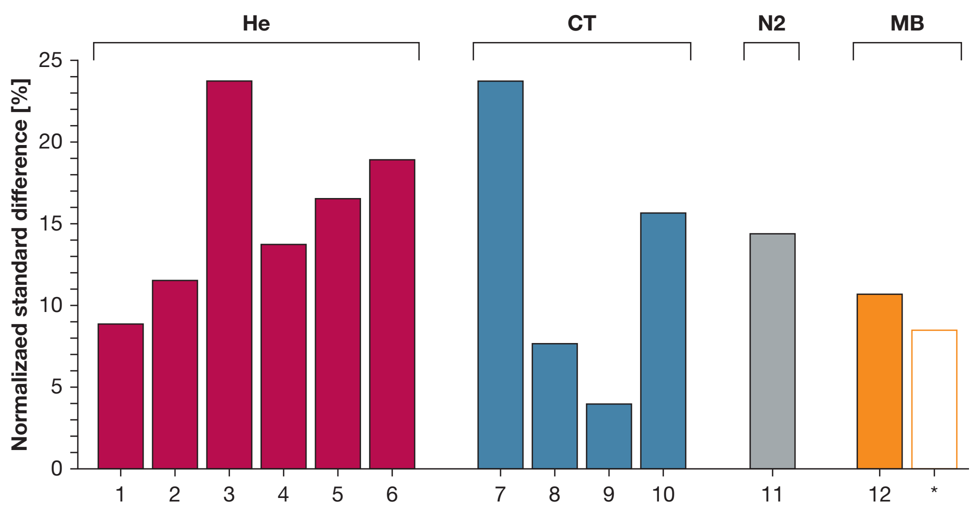
Participants

The study population comprised three groups of adult participants (age, ≥ 18 years): (1) healthy participants, (2) patients with airflow obstruction (FEV_1 to FVC, < 0.70), and (3) patients with restrictive ventilatory disorders. Disease severity varied and was defined based on ATS/ERS guidelines.[1] Participants were recruited either from referrals to the lung function laboratory, from referrals to the pulmonary outpatient clinic, or both. Healthy participants were recruited either by word of mouth (Vermont site) or from among healthy individuals who were referred for pulmonary function testing (other sites). Each site used different plethysmography devices, allowing analysis for site and device dependence. Each site's research ethics committee approved the study.

All participants provided informed consent and were capable of following instructions. Pregnant women were excluded. Healthy participants were recruited based on: (1) no smoking history (< 5 pack-years), (2) BMI < 30 kg/m^2, (3) absence of wheeze, (4) absence of respiratory symptoms (breathlessness, cough, or sputum), (5) no history of asthma or response to either bronchodilator or methacholine, and (6) absence of recent respiratory tract infection (within 6 months).

Study Protocol

Participants first performed spirometry using either the plethysmograph or the MiniBox spirometer. Each participant's year of birth was used to assign which device to use first; plethysmography was used first in individuals born in an even year, and the MiniBox was used first in individuals born in an odd year. Testing was performed by pulmonary function technicians, research coordinators (all with advanced degrees including nursing, registered pulmonary function technologist certification, and baccalaureate degrees), or both who specifically were trained to perform both spirometry and plethysmography testing procedures. All measurements conformed with ATS/ERS guidelines.[17,18] For slow



Author	N	Plethysmography devices
1. Andersson et al[4]	62	Body Box Ohio
2. Cliff et al[5]	38	PK Morgan
3. Kilburn et al[6]	16	Cardiopulmonary Instruments
4. Paré et al[7]	20	Collins
5. Rodenstein & Stanescu[3]	24	Home constructed
6. O'Donnell et al[2]	132	Collins, Jaeger, PK Morgan
7. Garfield et al[8]	59	Vmax, CareFusion
8. Matsumoto et al[9]	118	Medical Graphics
9. Tantucci et al[10]	27	Biomedin
10. O'Donnell et al[2]	132	Collins, Jaeger, PK Morgan
11. Cliff et al[5]	38	PK Morgan
12. Fredberg et al[11]	434	ZAN, MedGraphics, Jaeger, CareFusion
* Current study	266	MCG Elite, Vmax, Master Screen, Medisoft

Figure 1 – *Bar graph showing the NSD for TLC data when body plethysmography was compared with those obtained using He, N2, CT scanning, and the MB systems. Also shown are the number of participants (N) and the plethysmography devices in each study. Of note, the NSDs were calculated for these published studies using the identical definition that was applied in the present study. He = helium dilution; MB = MiniBox; N2 = nitrogen washout; NSD = normalized root mean square difference; TLC = total lung capacity.*

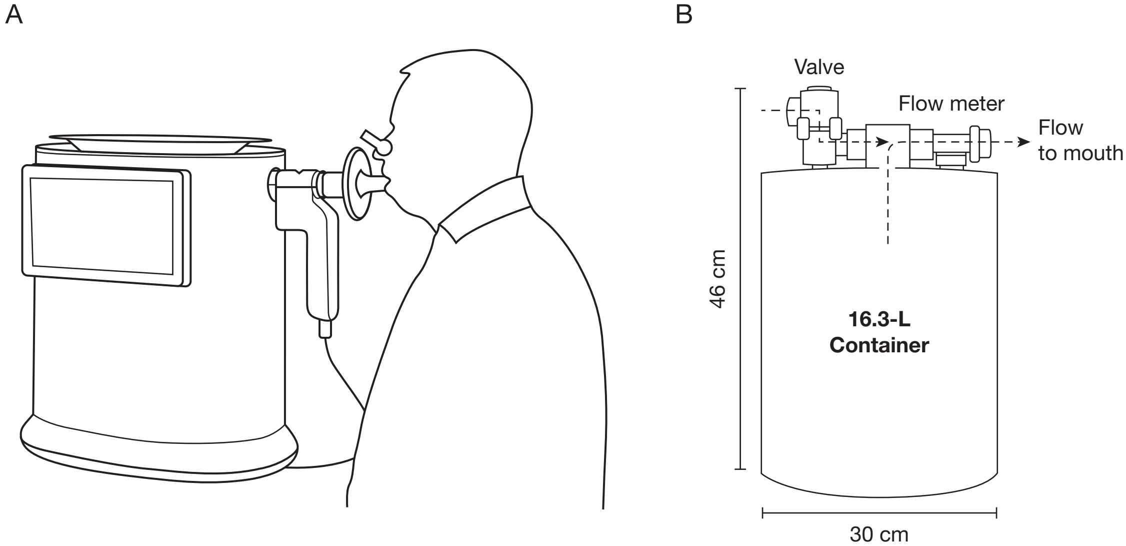


Figure 2 – *Illustrations of MiniBox device. A, Patient breathing through mouthpiece of the MiniBox, with the spirometer and the tablet computer (both detachable) attached to the device. B, Schematic of the device setup and airflow during the inhalation phase. When the valve is closed, air flows from the rigid container to the lungs, causing a pressure drop in the container.*

vital capacity and FVC, additional maneuvers were performed until the two highest values were within 0.15 L and 5% of each other.

Body Plethysmography

Body plethysmography devices varied by center (Barcelona, Medisoft, BodyBox, Sorinnes, Belgium; Vermont and Verona, MGC Diagnostics Elite, body plethysmograph, Saint Paul, MN; New York, Vyaire Vmax Encore, Yorba Linda, CA and Enschede Vyaire Master Screen (MasterScreen PFT System, Yorba Linda, CA). Measurements conformed with ATS/ERS guidelines.[18] Participants panted at FRC against a closed valve to measure thoracic gas volume, and then inhaled to full inspiration followed by slow exhalation to full expiration to calculate TLC and RV, respectively. Multiple thoracic gas volume measurements were obtained until repeatability met ATS/ERS guidelines.[18]

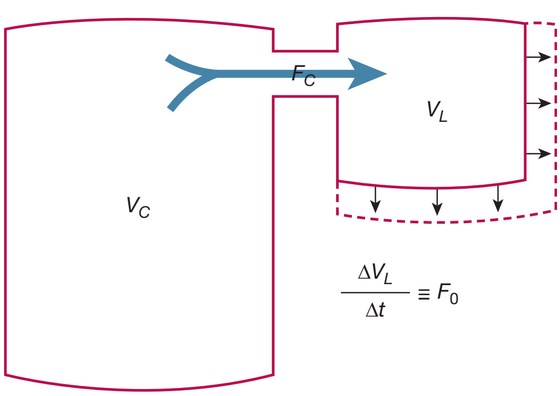


Figure 3 – *Diagram showing the physical basis of the MiniBox method. A rigid container of volume V_C is connected to a container of volume V_L, which expands at the rate F_0. Here V_L mimics the lungs and V_C mimics the volume of the MiniBox device. The rarefication of gas in the expanding system induces a flow of gas F_C from the rigid container to the expanding container.*

MiniBox Device

The MiniBox device comprises a detachable spirometer, a rigid container and tubing (total, 16.3 L), and a rapidly closing valve (< 15 msec) (Fig 2). The spirometer consists of two differential manometers with low and high dynamic ranges of ± 1.25 and 7 L/s, respectively. MiniBox flowmeter calibration was performed daily using a 3-L syringe.[17]

With a nose clip in place, each participant sat upright and breathed through a disposable bacterial and viral filter. TLC_{MB} was measured up to three times, with a final result provided without operator intervention based on automatic postprocessing of the flow data. The measurement was deemed acceptable if the slow vital capacity measured during the TLC_{MB} maneuver was within 0.15 L and 10% of the slow vital capacity measured by spirometry.

Description of the Novel Method

Physical Basis of MiniBox Method: As in traditional plethysmography, the MiniBox method is based on noninvasive measurements of gas pressures and flows.[11,19] Although the final determination of TLC by the MiniBox method depends on empirical adjustments that remain proprietary, the selection of measured parameters is guided by a physical model, as described below and in e-Appendix 1. The physical model derived here (Equation S17, E-Appendix 1) provides insights regarding the relationship of lung volumes to the flow interruption signal.

Figure 3 shows an idealized representation of the lung and MiniBox system during inhalation of air through the device with the valve closed (Fig 2A). The idealized system consists of a container with fixed volume, V_C, connected to a smaller container with volume V_L that expands at a rate of F_0. The fixed-volume container represents the MiniBox rigid container, the expanding volume represents the expanding lungs, and the flow in the tube connecting the two containers (F_C) caused by evacuation of air from the rigid container represents flow at the mouth (Fig 2B). In a closed system, the ratio of the expanding and rigid volumes (V_L to V_C) is in proportion to the ratio of the expansion and evacuation rates (F_0 to F_C) and to some measure of total system impedance. Further, the lungs are characterized by isothermal conditions (ie, Boyle's law), whereas

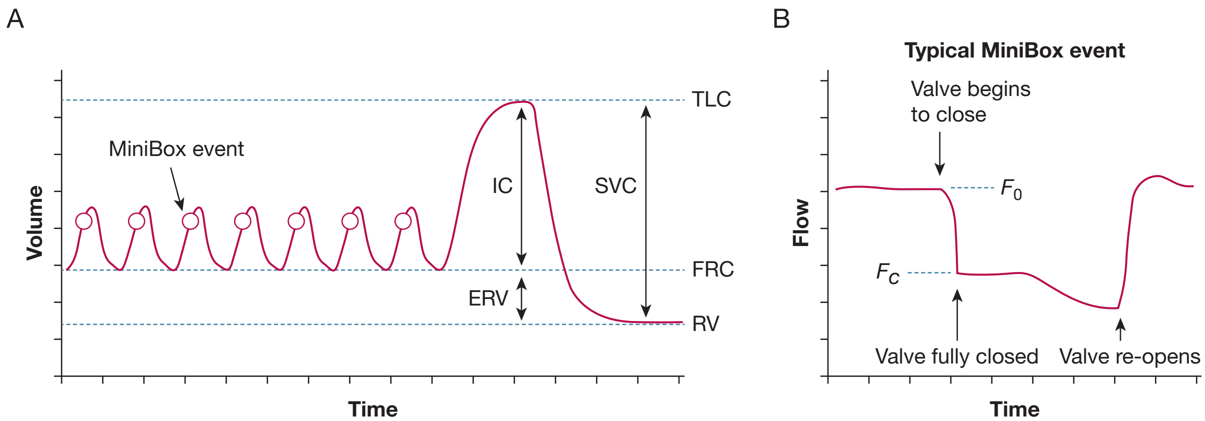


Figure 4 – *Graphs showing breathing maneuvers and events of the MiniBox. A, Minibox events (circles) are triggered toward the end of the inhalation phase of normal breathing. B, Airflow during a typical MiniBox event. The flow levels F_0 and F_C are calculated by averaging (over intervals of approximately 20 ms) the flow just before the valve begins to close and right after the valve fully closes. ERV = expiratory reserve volume; FRC = functional residual capacity; IC = inspiratory capacity; RV = residual volume; SVC = slow vital capacity; TLC = total lung capacity; VC = vital capacity.*

variations in the container are approximately adiabatic (ie, internal energy is conserved). As shown in e-Appendix 1, combining these constraints yields the following model for predicting V_L:

$$V_L = V_C\phi\left(\frac{F_0}{F_C}, \alpha, R\right), \quad (1)$$

where ϕ is a function of the unitless parameters F_0 to F_C, α, and R. The parameter α is a measure of the ratio of specific heats of the lung and container, which describes differences between the thermodynamic conditions in the lungs and container. The parameter R is a measure of the lung-container system impedance. The physical parameters α and R, as well as the specific functional form of ϕ, are determined empirically using plethysmography measurements of lung volume as reference.[11,19] More details on the physical assumptions of the method and on the derivation of Equation 1 are provided in e-Appendix 1.

MiniBox Measurement of TLC: TLC_{MB} is calculated from physiologic parameters together with a series of short flow interruption events (termed *MiniBox events*) activated during the inhalation phase of normal breathing. e-Appendix 1 provides a detailed description of how the parameters in Equation 1 are determined during these MiniBox valve closure events. Briefly, the MiniBox maneuver is shown in Figure 4A. Flow is measured continuously at the mouth, thus tracking relative changes in lung volume. The participant first breathes normally ($\geq$ 3 breathing cycles) to train the device to identify breathing cycles and to acclimatize themselves to the device. After these initial breathing cycles, the device automatically triggers MiniBox events after mid inspiration of each breathing cycle. After a minimum of six such events, the participant performs maximum inspiration to TLC, followed by exhalation to RV. The MiniBox events consist of rapid closure of the valve, after which air is evacuated from the rigid container by the lungs, followed by rapid reopening of the valve (after approximately 70 ms) and return to uninterrupted breathing (Fig 4B). For each event, TLC is calculated as the instantaneous lung volume during the event plus the volume inhaled to TLC; the average of these data is reported as the final TLC_{MB}.

Relationship of MiniBox Method to Plethysmography: Both traditional plethysmographic methods and the MiniBox method rely on measurements of pressures and flows. For that reason and others, the tendency of plethysmography to overestimate TLC in patients with chronic obstructive lung disease[5] is expected also to exist with the MiniBox. Similarly, TLC_{MB} is susceptible to errors resulting from nasal and oral air leaks.[20]

Statistical Analysis

The correlation between the two methods of measuring TLC was calculated using a two-tailed t test, with a significance level of 5% (P < .05). The difference between groups of patients was assessed by calculating the difference between TLC_{MB} and TLC_{Pleth} for each patient and using single-factor analysis of variance for comparing the groups of patients (F test between groups), with a 5% significance level.

The two methods are compared using Bland-Altman and identity plots. The comparison between techniques for measurement of TLC is summarized using the NSD, defined as:

$$\mathrm{NSD} = \frac{\mathrm{std}\ (TLC_{Pleth} - TLC_{MB})}{mean\ (TLC_{Pleth} + TLC_{MB})\ /\ 2}. \quad (2)$$

To account for inherent uncertainty in both TLC_{MB} and TLC_{Pleth} measurements, the linear dependence of TLC_{Pleth} values on TLC_{MB} values was calculated using a Deming regression,[21] with assumed equal uncertainty in both TLC_{MB} and TLC_{Pleth}. The 95% CIs for the slope were calculated as jackknife estimates as follows:

$$\pm 1.96SE\sqrt{1+\frac{1}{N}+\frac{(TLC_{MB} - \mathrm{mean}(TLC_{MB}))^2}{\sum(TLC_{MB} - \mathrm{mean}(TLC_{MB}))^2}}, \quad (3)$$

where SE is the standard error of the regression and N is the sample size.

The sample size of the present study was based on detection of differences in TLC_{MB} vs TLC_{Pleth} of 0.5 L assuming an SD of 1 L for TLC measurements in the study population. The initial intent was to recruit a total of 150 participants to achieve a power of 0.8. Considering the final size of the study population (N = 266 after addition of the Verona site) and the observed SD for TLC measurements of 0.48 L, the study had a power of 0.92 for detecting a difference in TLC measurements between devices of 0.1 L.

TABLE 1] Anthropometric Data of the Study Population

Variable	No.	Male Sex	Female Sex	Age, y	Height, cm	Weight, kg	BMI, kg/m^2
All	266	130	136	61 ± 17	169 ± 10	75 ± 18	26 ± 5
Healthy	36	14	22	38 ± 14	170 ± 11	74 ± 15	25 ± 5
Obstructed							
All	197	95	102	64 ± 15	168 ± 10	74 ± 17	26 ± 5
Mild	110	56	54	62 ± 15	169 ± 10	74 ± 18	26 ± 5
Moderate	42	17	25	67 ± 16	167 ± 9	74 ± 16	27 ± 5
Severe	45	22	23	64 ± 13	169 ± 11	73 ± 16	25 ± 4
Restricted							
All	33	21	12	66 ± 15	171 ± 9	85 ± 29	29 ± 9
Mild	23	15	8	66 ± 15	173 ± 8	83 ± 16	28 ± 5
Moderate and severe	10	6	4	65 ± 15	170 ± 10	90 ± 48	30 ± 14

Data are presented as No. or mean ± SD.

Results

Anthropometric data of the participants are shown in Table 1. A total of 266 participants were enrolled. No significant differences were found in height, sex, or age distributions among the five medical centers (with the exception of healthy participants who were significantly younger than patients). Because only one patient in the restricted group had severe obstruction, this patient's data were incorporated into a moderate and severe group.

No statistically significant differences were found in the distributions of TLC discrepancies in each center. Similarly, a leave-one-out cross-validation test showed that the overall results were not sensitive to a specific center (weighted NSD, 8.3% for cross-validation test vs 8.5% for total population). Because of the low number of participants in each severity group per center, a similar sensitivity analysis per severity was not statistically revealing. Therefore, we proceeded with a severity analysis of the overall population.

Overall Study Population

A Bland-Altman plot of all participants is shown in Figure 5A (overall NSD, 8.5%; mean discrepancy, –0.05 L). The relationship between TLC_{Pleth} and TLC_{MB} values is shown in Figure 5B. A tight correlation was found between TLC_{Pleth} and TLC_{MB} ($R^2 = 0.89$) with a slope of 1.024, that is, no clinically meaningful difference was found from the identity line. The measured TLC differed by ≥ 1 L between the techniques in 5.6% of the total study population (n = 15), compatible with expectations for a Gaussian distribution given that the 95% CI of the discrepancy between TLC measurement techniques was 0.96 L. These participants with a ≥ 1-L discrepancy in TLC measurements did not differ from the remaining study population with respect to age, sex, BMI, study center, disease type, or severity of disease.

The data were analyzed to identify instances where the final diagnosis with respect to either restriction or obstruction differed by device. First, data were analyzed from the 15 participants who demonstrated a > 1-L difference between TLC measurements. In all but one of these individuals, both the TLC_{Pleth} and TLC_{MB} agreed with respect to the diagnosis of obstruction or restriction. When the entire study population was analyzed, the final physiologic diagnosis differed by device in a total of 14 participants (five healthy participants, six obstructed participants, and three restricted participants). In 10 of these people, the TLC_{MB} characterized the physiologic pattern in accord with the clinical diagnosis.

Effect of Disease Type and Severity

A summary of results by disease type and severity is shown in Table 2. The NSD between the TLC_{Pleth} and TLC_{MB} measurements was 7.0% in healthy participants. For patients with obstructive disease (n = 197), the NSD ranged from 7.7% to 9.1%, depending on disease severity. The agreement between TLC_{MB} and TLC_{Pleth} measurements was similar in groups with mild and moderate obstruction with similar 95% CIs for TLC discrepancies (mild, –0.91 to 0.39 L; moderate, –0.72 to 0.51 L). The TLC_{MB} measurements for the group with severe obstruction were less than the TLC_{Pleth} measurements by an average of 0.12 L with a 40% larger spread of discrepancies (–0.82 to

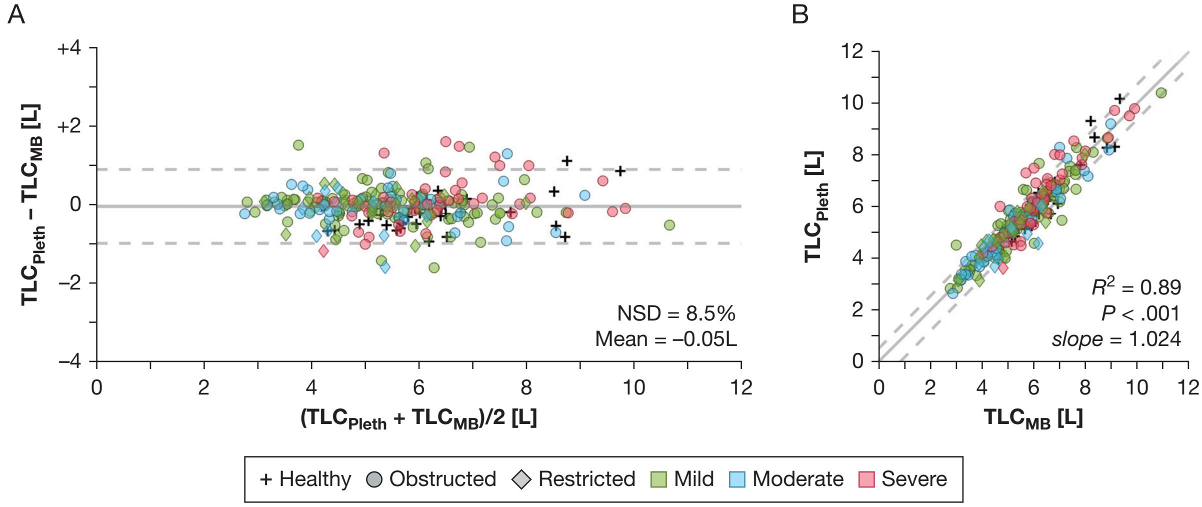


Figure 5 – *Comparison of* TLC_{MB} *and* TLC_{Pleth} *for all participants in the study. A, Bland-Altman plot in which the mean discrepancy and deviations of 1.96 SD from it are shown in solid and dashed lines, respectively. The NSD and the mean discrepancy are shown. B, Identity plot showing* TLC_{Pleth} *vs* TLC_{MB}*. Solid line is the identity line. Dashed lines show 95% CIs about a linear Deming regression.*[23] *The* R^2*, P value, and slope (1.024* $\pm$ *0.003) of the linear Deming regression also are shown. NSD = normalized root mean square difference;* TLC_{MB} *= total lung capacity measured by the MinBox;* TLC_{Pleth} *= total lung capacity measured by plethysmography.*

1.14 L), which may reflect increased heterogeneity of airway disease in these participants with severe airflow limitation; these differences were not statistically significant (P = .19).

Figure 6 shows a Bland-Altman plot and the relationship between TLC_{Pleth} and TLC_{MB} values in the obstructed patients. The NSD between TLC_{Pleth} and TLC_{MB} was 8.3%, with a mean discrepancy of 0.01 L. Similar to findings in the overall population, a high correlation between TLC_{Pleth} and TLC_{MB} values of obstructed patients was found ($R^2 = 0.90$) with a slope of 1.009, that is, no clinically meaningful difference from the identity line was found.

Figure 7 shows a Bland-Altman plot and the relationship between TLC_{Pleth} and TLC_{MB} values in participants with restriction. The overall NSD between TLC_{Pleth} and TLC_{MB} was 10.3%, with a mean discrepancy of –0.18 L. In the combined moderate and severe group, the NSD was higher at 13.9% (as might be expected because of lower TLC). For all participants with restriction, a high correlation between TLC_{Pleth} and TLC_{MB} values was found ($R^2 = 0.75$), with a slope of 1.004, that is, no

TABLE 2] TLC Data Obtained by Plethysmography and MiniBox Grouped by Participant Category and Disease Severity

Variable	FVC, L	FEV_1, L	TLC_{Pleth}, L	TLC_{MB}, L	NSD
All	3.2 ± 1.2	2.4 ± 1.0	5.6 ± 1.4	5.7 ± 1.4	8.5
Healthy	4.3 ± 1.0	3.5 ± 0.7	6.1 ± 1.4	6.3 ± 1.2	7.0
Obstructed					
All	3.0 ± 1.1	2.2 ± 1.0	5.7 ± 1.5	5.6 ± 1.5	8.3
Mild	3.3 ± 1.1	2.5 ± 0.8	5.5 ± 1.3	5.5 ± 1.4	7.9
Moderate	2.6 ± 1.0	1.8 ± 0.8	5.3 ± 1.5	5.3 ± 1.6	7.7
Severe	2.8 ± 1.1	1.7 ± 1.0	6.5 ± 1.5	6.3 ± 1.3	9.1
Restricted					
All	2.9 ± 1.0	2.3 ± 0.9	4.9 ± 1.0	5.1 ± 1.0	10.3
Mild	3.0 ± 1.1	2.4 ± 1.0	5.2 ± 1.1	5.2 ± 1.1	7.8
Moderate and severe	2.7 ± 0.7	2.1 ± 0.6	4.4 ± 0.8	5.0 ± 0.9	13.9

Data are presented as mean ± SD or percentage. NSD = normalized root mean square difference; TLC = total lung capacity; TLC_{MB} = total lung capacity by MiniBox; TLC_{Pleth} = total lung capacity by body plethysmography.

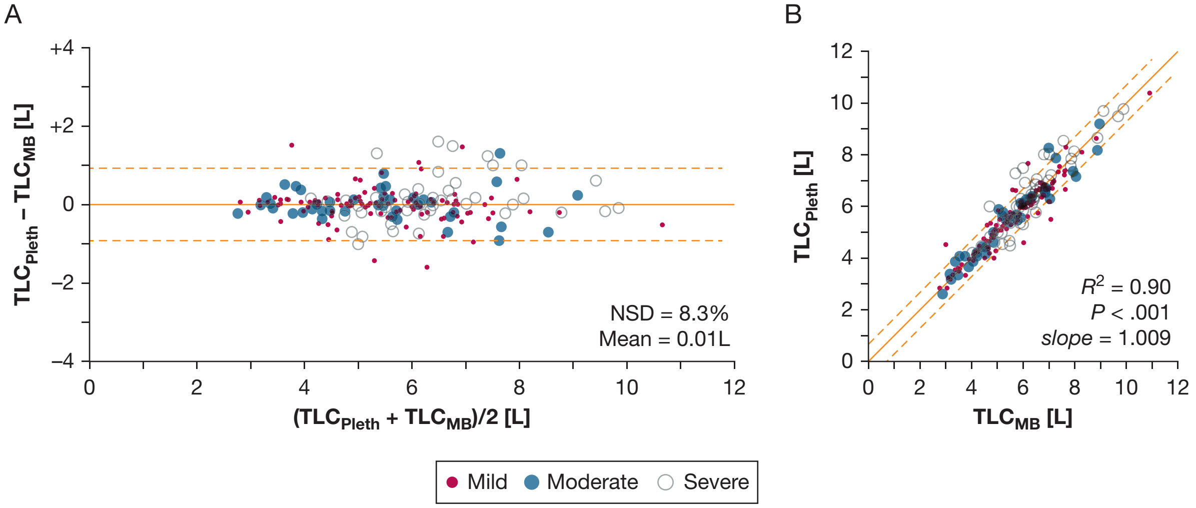


Figure 6 – *Comparison of TLC_{MB} and TLC_{Pleth} for the participants with obstructive dysfunction in the study. A, Bland-Altman plot in which the mean discrepancy and deviations of 1.96 SD from it are shown in solid and dashed lines, respectively. The NSD and the mean discrepancy are shown. B, Identity plot showing TLC_{Pleth} vs TLC_{MB}. Solid line is the identity line. Dashed lines show 95% CIs about a linear Deming regression. The R^2, P value, and slope (1.009 ± 0.003) of the linear Deming regression also are shown. NSD = normalized root mean square difference; TLC_{MB} = total lung capacity measured by the MinBox; TLC_{Pleth} = total lung capacity measured by plethysmography.*

clinically meaningful difference from the identity line was found.

Discussion

Lung volume measurement is an integral part of standard physiologic assessment.[1] Knowledge of lung volumes is important in obstructive lung disease, particularly in the evaluation of hyperinflation, which is an independent predictor of all-cause and respiratory mortality.[22] FRC frequently is increased in obstructive lung disease, and TLC can increase as well, especially in patients with severe emphysema with large bullae. Increased FRC is associated with expiratory flow limitation during tidal breathing and during exercise, with resultant increase in the elastic work of breathing. Measurement of lung volumes also is essential for the evaluation of restrictive respiratory diseases.[23] Guidelines and equipment recommendations for determination of lung volumes are derived using physiologic and physics-based principles, but even the

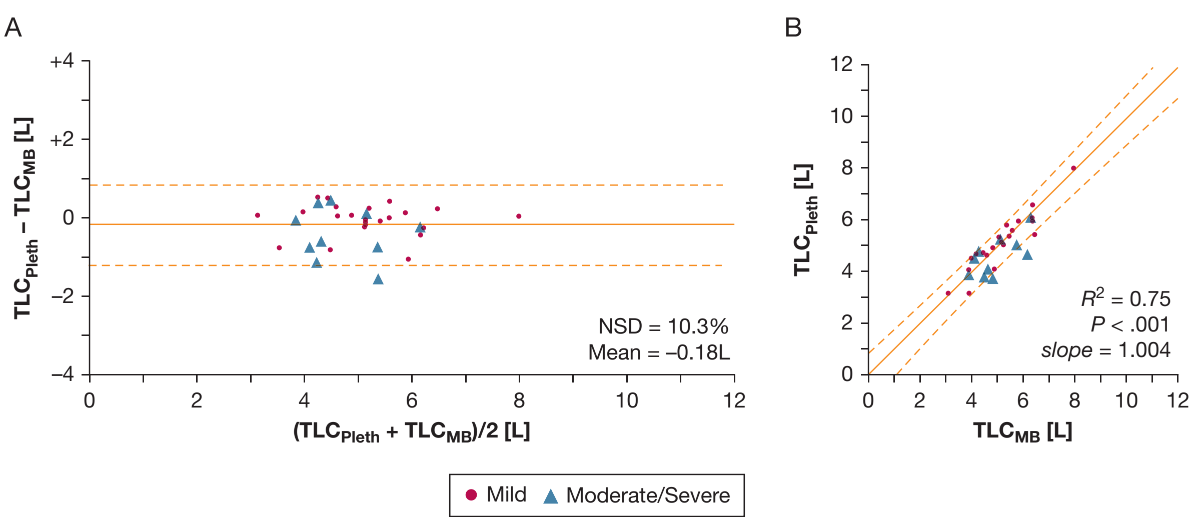


Figure 7 – *Comparison of TLC_{MB} and TLC_{Pleth} for the participants with restrictive dysfunction in the study. A, Bland-Altman plot in which the mean discrepancy and deviations of 1.96 SD from it are shown in solid and dashed lines, respectively. The NSD and the mean discrepancy are shown. B, Identity plot showing TLC_{Pleth} vs TLC_{MB}. Solid line is the identity line. Dashed lines show 95% CIs about a linear Deming regression. The R^2, P value, and slope (1.004 ± 0.031) of the linear Deming regression also are shown. NSD = normalized root mean square difference; TLC_{MB} = total lung capacity measured by the MinBox; TLC_{Pleth} = total lung capacity measured by plethysmography.*

accepted plethysmography gold standard does not ensure accuracy of the measurements in patients with respiratory disease.

The choice of technique to measure lung volumes depends on availability, cost, convenience, and accuracy. The ATS/ERS Consensus Statement identifies five methods: whole body plethysmography, helium dilution, nitrogen washout, CT scanning, and chest radiography.[18] The choice of an optimal system is easier in patients with pure restrictive disease because all techniques yield similar values. Accuracy becomes more difficult when assessing patients with a mixed restrictive-obstructive disorder and even more difficult in patients with obstructive airways disease. Among these standard methods, plethysmography is more accurate, but its use in the clinical setting may be limited by increased expense and difficulty obtaining measurements in patients with claustrophobia or individuals with physical disabilities. Nevertheless, plethysmography has become widely accepted globally, and therefore we chose it for comparison with the novel MiniBox measurement system.

Unlike plethysmography, determination of TLC by the MiniBox is based on inductive statistics and pattern recognition algorithms that are used to infer parameters of a mathematical model automatically. The present study demonstrated that this approach can be applied to a highly heterogeneous population of participants, yielding accurate and reproducible determination of TLC as compared with the current gold standard plethysmography technique.

Data reported here compare favorably with the only other multicenter study[2] that assessed TLC by a variety of techniques in a large number of healthy participants and respiratory patients. With permission from the authors,[2] we analyzed the raw data from the study comparing TLC_{Pleth} measurements with TLC measurements obtained by CT scanning (e-Fig 1) and by helium dilution (e-Fig 2). In our study, the NSD was 8.5% for TLC_{MB} vs TLC_{Pleth}, which compares favorably with an NSD of 11.8% and 16.6% for TLC measurements obtained by CT scanning and by helium dilution vs TLC_{Pleth} measurements, respectively.[2] Moreover, Figure 1 highlights that the agreement between the MiniBox and plethysmography methods is favorable as compared with other smaller studies that used numerous plethysmograph devices and numerous comparator techniques to assess TLC. These findings indicate that the MiniBox can derive absolute lung volumes precisely in a variety of participants and patients with diseases of varying severities.

Differences in measurement standards and in the manufacturer of plethysmography devices can be a potential source of discrepancy between TLC_{Pleth} and TLC_{MB} measurements. A study recently presented at the ERS International Congress compared five plethysmography devices from the same manufacturer and found systematic differences between measured lung volumes.[24] In the present study, we tested a wide spread of patients from European and US centers using four different plethysmography devices. Sensitivity analyses confirmed that the agreement between TLC_{MB} and TLC_{Pleth} measurements was equivalent across plethysmography devices.

Study limitations include a relatively small sample size, particularly for some patient subgroups (eg, restrictive dysfunction), a narrow age distribution for patients with disease, and a younger healthy group. Nevertheless, the discrepancies between TLC_{MB} and TLC_{Pleth} measurements were similar across disease subgroups. In addition, this study did not assess day-to-day variability of MiniBox measurements, although in a previous study, the day-to-day variability in 26 healthy participants (expressed as a coefficient of variation) was 1.6% for TLC_{MB} compared to 3.3% for TLC_{Pleth}.[11] Finally, isolated TLC_{MB} values differed by 1 L compared with TLC_{Pleth} values in healthy participants and patients with respiratory diseases, consistent with real-life experience and other published studies.[2-10] Importantly, in these people, the clinical diagnosis was aligned more closely to physiologic pattern based on TLC derived by the MiniBox than by plethysmography.

The National Heart, Lung and Blood Institute, ATS, and ERS have encouraged innovation in technologies to measure absolute lung volumes to attain improved accuracy, ease of use, and rapid assessment.[25] Rigorous testing is suggested to ensure that results are statistically comparable with standard techniques. In all subpopulations tested, the MiniBox performed in a manner that compared favorably with plethysmography. Moreover, the measurement of lung volumes is facilitated by reduced cost of the system as compared with plethysmography and by the ability to obtain data during tidal breathing without need for complex respiratory maneuvers. The MiniBox system is mobile and simpler to operate when compared with other lung volume measurement systems. In addition, the compact size of the MiniBox facilitates and speeds disinfection of the hardware between patients.[26-29] These considerations indicate that the MiniBox system should be considered a clinically reliable technique to obtain measurement of lung volumes that are equivalent to plethysmography values in both healthy participants as well as those with lung disease.

Acknowledgments

Author contributions: R. J. S. and K. I. B. designed the study. D. A. K., F. B., F. H. C. d. J., R. W. D. N., and K. I. B. oversaw the study in terms of patient recruitment and data recording. I. C. and O. A. had access to and analyzed all data. R. J. S., O. A., D. A. K., F. B., F. H. C. d. J., R. W. D. N., K. I. B., and J. J. F. contributed to the writing of the manuscript. All authors have seen and approved the final version of the manuscript and take responsibility for the integrity of the work from inception of the study to publication.

Financial/nonfinancial disclosures: The authors have reported to *CHEST* the following: R. J. S., O. A., K. I. B., and J. J. F. have a financial interest in PulmOne Advanced Medical Devices, Ltd. F. B. is a member of the scientific advisory board of Medical Graphics Corporation Diagnostics. None declared (R. W. D. N., D. A. K., F. H. C. d. J., I. C.).

Role of sponsors: The sponsor did not participate in the data analysis or drafting the contents of the manuscript.

Other contributions: The authors thank Robert Brown (Harvard University, Boston, MA) and Julian Solway (University of Chicago, Chicago, IL) for their input regarding the development and evaluation of the MiniBox system, C. R. O'Donnell for sharing the data shown in e-Figures 1 and 2, and Mr Pietro Longo.

Additional information: The e-Appendix and e-Figures can be found in the Supplemental Materials section of the online article.

References

1. Pellegrino R, Viegi G, Brusasco V, et al. Interpretative strategies for lung function tests. *Eur Respir J*. 2005;26(5):948-968.
2. O'Donnell CR, Bankier AA, Stiebellehner L, Reilly JJ, Brown R, Loring SH. Comparison of plethysmographic and helium dilution lung volumes: which is best for COPD? *Chest*. 2010;137(5):1108-1115.
3. Rodenstein DO, Stanescu DC. Reassessment of lung volume measurement by helium dilution and by body plethysmography in chronic air-flow obstruction. *Am Rev Respir Dis*. 1982;126(6):1040-1044.
4. Andersson LG, Ringqvist I, Walker A. Total lung capacity measured by body plethysmography and by the helium dilution method. A comparative study in different patient groups. *Clin Physiol*. 1988;8(2):113-119.
5. Cliff IJ, Evans AH, Pantin CF, Baldwin DR. Comparison of two new methods for the measurement of lung volumes with two standard methods. *Thorax*. 1999;54(4):329-333.
6. Kilburn KH, Miller A, Warshaw RH. Measuring lung volumes in advanced asbestosis: comparability of plethysmographic and radiographic versus helium rebreathing and single breath methods. *Respir Med*. 1993;87(2):115-120.
7. Paré PD, Wiggs BJ, Coppin CA. Errors in the measurement of total lung capacity in chronic obstructive lung disease. *Thorax*. 1983;38(6):468-471.
8. Garfield JL, Marchetti N, Gaughan JP, Steiner RM, Criner GJ. Total lung capacity by plethysmography and high-resolution computed tomography in COPD. *Int J Chron Obstruct Pulmon Dis*. 2012;7:119-126.
9. Matsumoto AJ, Bartholmai BJ, Wylam ME. Comparison of total lung capacity determined by plethysmography with computed tomographic segmentation using CALIPER. *J Thorac Imaging*. 2017;32(2):101-106.
10. Tantucci C, Bottone D, Borghesi A, Guerini M, Quadri F, Pini L. Methods for measuring lung volumes: is there a better one? *Respiration*. 2016;91(4):273-280.
11. Fredberg JJ, Cohen I, Adam O, et al. Determination of total lung capacity (TLC) without body plethysmography. *Am J Respir Crit Care Med*. 2014;189:A2933.
12. Dahlqvist M, Hedenstierna G. Lung volumes measured by helium dilution and by body plethysmography with mouth and oesophageal pressures: a comparative study in patients with lung disease. *Clin Physiol*. 1985;5(2):179-187.
13. Dorkin HL, Lutchen KR, Jackson AC. Human respiratory input impedance from 4 to 200 Hz: physiological and modeling considerations. *J Appl Physiol*. 1988;64(2):823-831.
14. Chalker RB, Celli BR, Habib RH, Jackson AC. Respiratory input impedance from 4 to 256 Hz in normals and chronic airflow obstruction: comparisons and correlations with spirometry. *Am Rev Respir Dis*. 1992;146(3):570-576.
15. Frey U, Suki B, Kraemer R, Jackson AC. Human respiratory input impedance between 32 and 800 Hz, measured by interrupter technique and forced oscillations. *J Appl Physiol*. 1997;82(3):1018-1023.
16. Flesch JD, Dine CJ. Lung volumes: measurement, clinical use, and coding. *Chest*. 2012;142(2):506-510.
17. Miller MR, Hankinson J, Brusasco V, et al. Standardisation of spirometry. *Eur Respir J*. 2005;26(2):319-338.
18. Wanger J, Clausen JL, Coates A, et al. Standardisation of the measurement of lung volumes. *Eur Respir J*. 2005;26(3):511-522.
19. Adam O, Cohen I, Yip W-K, et al. Total lung capacity without plethysmography. *bioRxiv*. 2018:395160. https://doi.org/10.1101/395160.
20. Peslin R. Body plethysmography. In: *Techniques in the Life Sciences: Respiratory Physiology*. 4th ed. County Clare, Ireland: Elsevier Scientific Publishers; 1984:1-26.
21. Deming WE. *Statistical Adjustment of Data*. New York: Dover Publications; 2011.
22. Casanova C, Cote C, de Torres JP, et al. Inspiratory-to-total lung capacity ratio predicts mortality in patients with chronic obstructive pulmonary disease. *Am J Respir Crit Care Med*. 2005;171(6):591-597.
23. Hughes JMB. Interpreting pulmonary function tests. *Breathe*. 2009;6(2):102-110.
24. De Soomer K, Adriaensen C, Leemans K, De Backer W, Oostveen E. Systematic differences in lung function outcomes in a new lung function lab. *Eur Respir J*. 2017;50(suppl 61):PA3003.
25. NHLBI workshop. Consensus statement on measurement of lung volumes in humans. www.thoracic.org/adobe/lungvolume.pdf. Date last updated December 30, 2003. Accessed March 23, 2021.
26. ERS Group 9.1. Lung function testing during COVID-19 pandemic and beyond. European Respiratory Society website. https://ers.app.box.com/s/zs1uu88wy51monr0ewd990itoz4tsn2h. Accessed September 10, 2020.
27. Irish Thoracic Society. Guidance on lung function testing: SARS COVID-19. Irish Thoracic Society website. https://irishthoracicsociety.com/wp-content/uploads/2020/03/Guideline-on-Lung-Function-Testing-V2-30.03.pdf. Accessed September 10, 2020.
28. McCormack MC, Kaminsky DA. Pulmonary function laboratories: advice regarding COVID-19. American Thoracic Society website. https://www.thoracic.org/professionals/clinical-resources/disease-related-resources/pulmonary-function-laboratories.php. Accessed September 10, 2020.
29. Demain JG. Spirometry during COVID-19. American Academy of Allergy, Asthma, and Immunology website. https://www.aaaai.org/ask-the-expert/spirometry. Accessed September 10, 2020.

PulmOne
PFT Outside The Box

12/15/2023

The purpose of this letter is to verify the uniqueness of the PulmOne MiniBox+.

The MiniBox+ is the only cabinless, portable device with full Pulmonary Function testing capabilities.

While other devices provide spirometry and diffusion capacity testing in a similar way, the cabinless and gasless plethysmography available through the Minibox+ is unique. It provides a patient experience with no sensation of claustrophobia and can test obese, wheelchair bound or non-ambulatory patients.

The MiniBox+ has been clinically validated in a groundbreaking, multi-center study. The results of that study were published in the CHEST magazine in February 2021 (https://doi.org/10.1016/j.chest.2021.01.052)

PulmOne USA Inc. is the sole manufacturer and provider of the MiniBox+ worldwide.

Respectfully,

Andreas Lex

Andreas Lex,
President, PulmOne USA

PulmOne USA Inc.

444 Madison Ave
New York, NY 10022

,
To Whom It May Concern:

Easy. Accurate. Available. Lung Function Testing

HL7 CONFORMANCE STATEMENT

PRODUCT DETAILS:
MiniBox+

MANUFACTURER:
PulmOne Advanced Medical Devices Ltd.
14 HaMelacha street
Ra'anana, 4366107 ISRAEL
US Tel: 1 844 PulmOne (785 6663)
Tel: +972 (77) 510 0938
info@pulm-one.com

STANDARDS:
HL7 standard v2

Table of Contents

Introduction

Overview

This document provides the requirements to interoperate with the PulmOne MiniBox+ using the Health Level Seven (HL7) standard.

The objective of this document is to provide a standard set of HIS/CIS/RIS requirements to effectively support and connect to the PulmOne MiniBox+.

Intended Audience

The user of this document is involved with system integration and/or software design. We assume that the reader is familiar with the terminology and concepts that are used in HL7 V2 standard. Readers not familiar with HL7 V2 terminology should first read the appropriate parts of the HL7 V2 standard itself, prior to reading this conformance statement.

Although the use of this conformance statement in conjunction with the HL7 V2 standard is intended to facilitate communication with PulmOne MiniBox+, it is not sufficient to guarantee, by itself the inter-operation of the connection between PulmOne MiniBox+ and the 3rd party HL7-based system.

The integration of any device into a system of interconnected devices goes beyond the scope of the HL7 V2 standard and this conformance statement when interoperability is desired. The responsibility for analyzing the applications requirements and developing a solution that integrates the PulmOne MiniBox+ with other vendors' systems is the user's responsibility.

Standard Version Control

The PulmOne MiniBox+ supports version 2.5.1 of the HL7 V2 protocol. The MiniBox+ has also been successfully integrated with HL7 servers using HL7 V2 versions 2.3 and 2.4. In general, the PulmOne MiniBox+ is capable of integrating with servers using higher versions of HL7 V2. In such cases contact the PulmOne service team for further advice.

Overview

Diagram

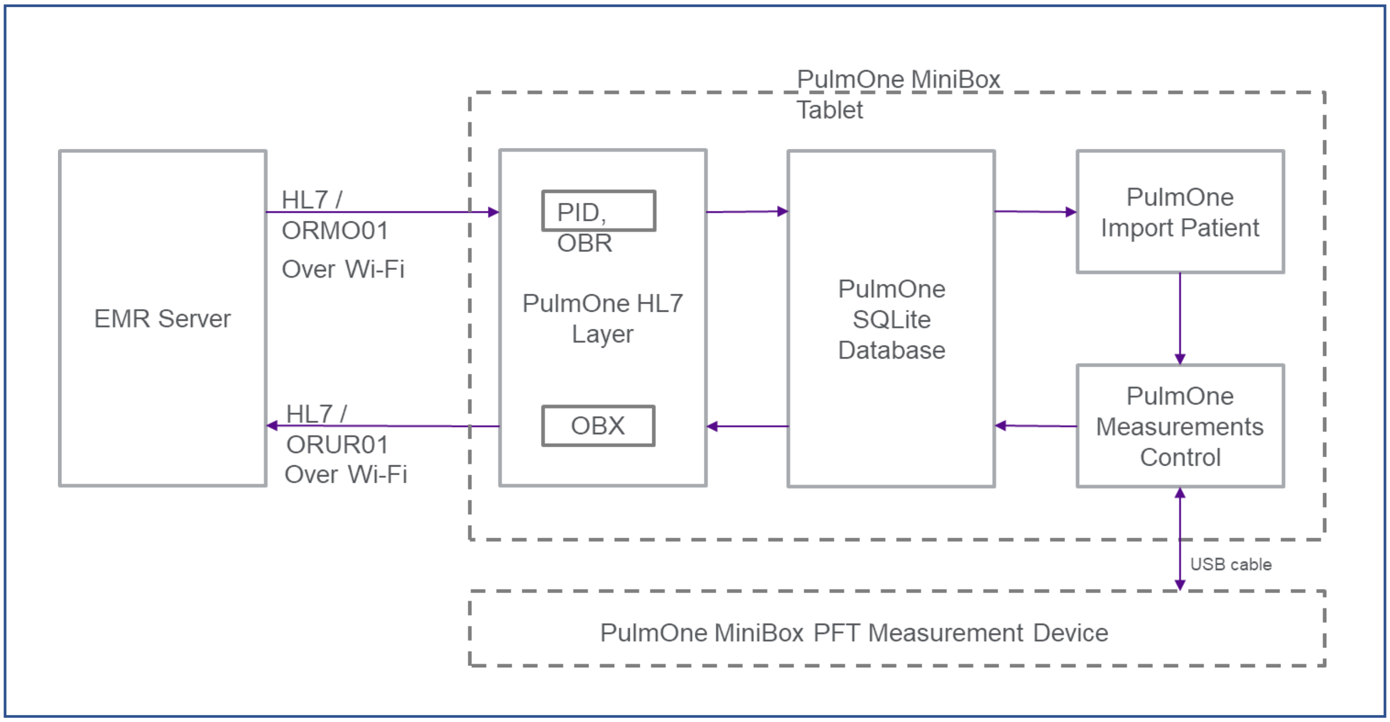


Figure 1: PulmOne MiniBox HL7 Integration

General

- The PulmOne MiniBox device is capable of working in either a standalone mode or alternatively, it can function as an HL7 client, communicating with an HL7 EMR server, using a subset of the HL7 protocol as describe in Figure 1.
- All the software and hardware required for the HL7 communications are built into the MiniBox tablet.
- The MiniBox receives the patients and visits information from the EMR server using the HL7 **ORMO01** message. The MiniBox supports both new visits as well as visit cancellations.
- The MiniBox sends the test results back to the EMR server using the HL7 **ORUR01** message. This message includes the measured values of all the discrete parameters that were selected by the user using the settings screen, as well as their predicted values, their lower and upper level of normal and their Z-Score.
- The ORUR01 message sent by the MiniBox also includes an encapsulated visit summary PDF report, encoded using Base64. Alternatively, the MiniBox can store the summary PDF report on a network drive, provided by the customer and include the name of the visit's summary PDF report file as part of the ORUR01 message.

Logical Architecture

PulmOne's MiniBox device enables Physicians to conduct pulmonary and diffusion testing. The device includes a measuring hardware unit (12 x 18 x 20 in) and a 12in tablet which runs the minibox software that controls the measuring unit. The software is implemented using Windows 10 operating system, .NET Framework, C#, WPF, MVVM.

- Presentation Layer: Includes a custom UI layer that the physician and patients interact with, using the tablet's 12 in touchscreen that includes a virtual keyboard.
- Application Layer: Communicates with the Measurement hardware unit, controls the measurements and validates the measurement results.
- Data Access & Integration Layer: Communicates with the EMR server using the HL7 protocol. Receives the patient demographic and required testing from the EMR server and sends the test results, as well as the Physician diagnostics to the EMR server.
- Data Layer: Stores the Measurement results on a local SQLite database that resides on the MiniBox tablet.

PulmOne MiniBox Tablet Configuration

PulmOne MiniBox Tablet Specifications
Tablet model – DELL Latitude 5175 or Microsoft Go 3
CPU: Intel® Core™ m5-6Y57 Processor (4M Cache, up to 2.80 GHz)
Memory: 4GB (1x4GB) DDR3L 1600MHz Memory
Storage: M.2 128GB SATA Class 20 Solid State Drive
Operating System – Windows 10 Professional

- No external Servers required
- Alternatively using a Jumper EZPAD PRO 6 tablet with similar specifications

Security Architecture

- The PulmOne MiniBox includes functionality for encryption/decryption of all confidential patient and test results data. The data is saved in an SQLite database that resides on an encrypted drive in the tablet itself.
  - Local Encryption method: Windows BitLocker
- The PulmOne MiniBox supports authorization module by user name and password.
  - The system enforces password rules when creating passwords (min length, at least one of each lower/upper case letters, digits and special characters).
  - The system locks the account after 5 login failures.
  - The system logs the user out automatically after 10 min of inactivity (configurable) and returns to the home screen.

HL7 Interface Description and requirements

- Connection Type: TCP/IP
- MiniBox Listening Port: 8080 (configurable)
- The Firewall (if any) needs to allow TCP/IP connection/messages from the IP address of the EMR server HL7 order interface to the MiniBox listening port.
- The Firewall (if any) needs to allow connections/messages from the IP address of the MiniBox to the EMR server results interface listening port.
- It is expected that the DHCP server assigns a static IP address for the MiniBox (and the EMR server) to prevent connections/messages being blocked after a power failure or network maintenance.

Detailed HL7 messages and segments description

Order Message (ORMO01)

The HL7 Order Message (ORMO01) from the HL7 EMR server to the PulmOne MiniBox should contain the following segments:

- Message Header (MSH)
- Patient Id (PID)
- Patient Visit (PV1)
- Common Order (ORC)
- Observation Request (OBR)

Fields marked in the following tables as Required, are parsed and used. Fields Marked as Optional, are parsed and echoed back in the ORUR01 message. Fields marked as Not Used are ignored.

Message Header (MSH)

Message Header (MSH)			
Position	**Element**	**Required?**	**Sample Content**
0	Message Name	Required	MSH
1	Field Separator	Required	|
2	Encoding Characters	Required	^~\&
3	Sending Application	Required	EMR Server Name
4	Sending Facility	Required	Facility Name
5	Receiving Application	Optional	MiniBox
6	Receiving Facility	Optional	PulmOne
7	Date Time of message	Required	20201105133018
8	Security	Not Used	
9	Message Type	Required	ORM^O01
10	Message Control	Optional	P
11	Processing Id	Not Used	
12	Version Id	Not Used	
13 to 18		Not Used	
19	Principle Language of Message(*)	Optional	SPIRO

(*) Starting in version 3.3.0 of the MiniBox+ application, Not Used in previous versions

Patient Id (PID)

Patient Identifier (PID)			
Position	**Element**	**Required?**	**Sample Content**
0	Message Name	Required	PID
1	Set Id	Not Used	
2	Patient Id-External	Not Used	
3	Patient Id-Internal	Required	52068866
4	Alternate Patient Id	Not Used	
5	Patient Name	Required	LASTNAME^FIRSTNAME
6	Mother Maiden Name	Not Used	
7	Date of Birth	Required	19760821
8	Sex	Required	M or F
9	Patient Alias	Not Used	
10	Ethnic Group	Optional	Caucasian
11	Patient Address	Optional	123 FAKE ST^^NEW YORK^NY^10010^US
12	County Code	Not Used	
13	Phone Number-Home	Optional	(212)555-5555
14	Phone Number-Business	Not Used	
15	Language	Not Used	
16	Marital Status	Not Used	
17	Religion	Not Used	
18	Patient Account Number	Optional	496677136
19	Social Security Number	Optional	123-45-6789
20	Driver's License	Not Used	
>=21		Not Used	

Patient Visit (PV1)

Patient Visit (PV1)			
Position	**Element**	**Required?**	**Sample Content**
0	Message Name	Required	PV1
1	Set Id	Not Used	
2	Patient Class	Not Used	
3	Assigned Patient Location	Not Used	
4	Admission Type	Not Used	
5	Pre-Admit Number	Not Used	
6	Prior Patient Location	Not Used	
7	Attending Doctor	Optional	FAKE^NAME1
8	Referring Doctor	Optional	FAKE^NAME2
9	Consulting Doctor	Not Used	
10	Hospital Service	Not Used	
11	Temporary Location	Not Used	
12	Pre-Admit Test Indicator	Not Used	
13	Re Admission Indicator	Not Used	
14	Admit Source	Not Used	
15	Ambulatory Status	Not Used	
16	VIP Indicators	Not Used	
17	Admitting Doctor	Not Used	
18	Patient Type	Not Used	
19	Visit Number	Optional	22228615
20	Financial Class	Not Used	
21	Charge Price Indicator	Not Used	
22	Courtesy Code	Not Used	
23	Credit Rating	Not Used	
24	Contract Code	Not Used	
25	Contract Effective Date	Not Used	
26	Contract Amount	Not Used	
27	Contract Period	Not Used	
28	Interest Code	Not Used	
29	Transfer To Bad Debt Code	Not Used	
30	Transfer To Bad Debt Date	Not Used	
31	Bad Debt Agency Code	Not Used	
32	Bad Debt Transfer Amount	Not Used	
33	Bad Debt Recovery Amount	Not Used	

34	Delete Account Indicator	Not Used	
35	Delete Account Date	Not Used	
36	Discharge Disposition	Not Used	
37	Discharge To Location	Not Used	
38	Diet Type	Not Used	
39	Servicing Facility	Not Used	
40	Bed Status	Not Used	
41	Account Status	Not Used	
42	Pending Location	Not Used	
43	Prior Temporary Location	Not Used	
44	Admit Date Time	Not Used	
45	Discharge Date Time	Not Used	
46	Current Patient Balance	Not Used	
47	Total Charges	Not Used	
48	Total Adjustments	Not Used	
49	Total Payments	Not Used	
>=50		Not Used	

Common Order (ORC)

Common Order (ORC)			
Position	**Element**	**Required?**	**Sample Content**
0	Message Name	Required	ORC
1	Order Control	Required	NW (new order) or CA (order cancellation)
2	Placer Order Number	Not Used	
3	Filler Order Number	Not Used	
4	Placer Group Number	Not Used	
5	Order Status	Not Used	
6	Response Flag	Not Used	
7	Timing Quantity	Not Used	
8	Parent	Not Used	
9	Date Time of Transaction	Not Used	
10	Entered Ny	Not Used	
11	Verified By	Not Used	
12	Ordering Provider	Not Used	
>=13		Not Used	

Observation Request (OBR)

Observation Request (OBR)			
Position	**Element**	**Required?**	**Sample Content**
0	Message Name	Required	OBR
1	Set Id	Not Used	
2	Placer Order Number	Optional	73289718
3	Filler Order Number	Not Used	
4	Universal Service Id	Required	PFT13^PULMONARY FUNCTION TEST
5	Priority	Not Used	
6	Requested Date Time	Not Used	
7	Observation Date Time	Not Used	
8	Observation End date Time	Not Used	
9	Collection Volume	Not Used	
10	Collection Id	Not Used	
11	Specimen Action Code	Not Used	
12	Danger Code	Not Used	
13	Relevant Clinical Info	Not Used	
14	Specimen Received Date Time	Not Used	
15	Specimen Source	Not Used	
16	Ordering Provider	Optional	NAME
17	Order Callback Phone number	Not Used	
18	Placers Field 1	Not Used	
19	Placers Field 2	Not Used	
20	Filler Field 1	Not Used	
21	Filler Field 2	Not Used	
22	Results Report Status Change dt	Not Used	
23	Charge To Practice	Not Used	
24	Diagnostic Service Sect Id	Not Used	
25	Result Status	Not Used	
26	Linked Results	Not Used	
27	Quantity timing	Not Used	
28	Results Copies To	Not Used	
29	Parent Accession Number	Not Used	
30	Transportation Mode	Not Used	
31	Reason for Study	Not Used	
32	Print Result Interpreter	Not Used	

33	Assist Result Interpreter	Not Used	
34	Technician	Not Used	
35	Transcriptionist	Not Used	
36	Scheduled Date Time	Not Used	
>=37		Not Used	

Other valid options for OBR4:

PulmOneSPR^Pulm One Spirometry ➔ for FVC only

PulmOneSVC^Pulm One Slow Vital ➔ for SVC only

PulmOneMVV^Pulm One Voluntary ➔ for MVV only

PFT13^COMPLETE PULMONARY FUNCTION TEST ➔ FVC, LVM and DLCO

PFT13^PULMONARY FUNCTION TEST ➔ FVC, LVM and DLCO

PFT Full ➔ FVC, LVM and DLCO

Sample ORMO01 message from an HL7 EMR server to PulmOne MiniBox

```
MSH|^~\&|EPIC|FakeFacility|||20201105133018||ORM^O01|22|T|2.3
PID|||52068865||LASTNAME^FIRST ||19760821|F|||123 FAKE ST^^NEW
YORK^NY^10010^US^^^60||(212)555-5555|||||496677135
PV1||||||||||||||||||||||||||||||||||||||||||||||||
ORC|NW|162406141|||||20201105133000^R||20201105133017|FAKE^
^JOSSIE^^||
OBR||162406141||PFT13^COMPLETE PULMONARY FUNCTION
TEST||20201105133000|20201105133017|||||||||
```

Observation Result Unsolicited (ORUR01)

The HL7 Observation Result Unsolicited (ORUR01) sent from the PulmOne MiniBox to the HL7 EMR server would contain the following segments:

- Message Header (MSH)
- Patient Id (PID)
- Patient Visit (PV1)
- Common Order (ORC)
- Observation Request (OBR)
- Observation Result (OBX)

Fields marked as Required in the following tables are always included. Fields Marked as Optional are parsed from the ORUR01 message and echoed back. Fields marked as Not Used are not included in the message.

Message Header (MSH)

Message Header (MSH)			
Position	**Element**	**Required?**	**Sample Content**
0	Message Name	Required	MSH
1	Field Separator	Required	|
2	Encoding Characters	Required	^~\&
3	Sending Application	Required	EMR Server Name
4	Sending Facility	Required	Facility Name
5	Receiving Application	Required	MiniBox
6	Receiving Facility	Required	PulmOne
7	Date Time of message	Required	20201105133018
8	Security	Not Used	
9	Message Type	Required	ORU^R01
10	Message Control	Optional	P
11	Processing Id	Not Used	
12	Version Id	Not Used	
13	Sequence Number	Not Used	
14	Continuation Pointer	Not Used	
15	Accept Ack Type	Not Used	
16	Application Ack type	Not Used	
17	Country Code	Not Used	
18	Character Set	Not Used	
19	Principal Language of Msg(*)	Optional	SPIRO

20	Alt Character Set Handling	Not Used	
21	Msg Profile Id	Required	LRI_NG_RN_Profile^HL7^2.16.840.1.113883.9.20^ISO
>=22		Not Used	

(*) Starting in version 3.3.0 of the MiniBox+ application, Not Used in previous versions

Patient Id (PID)

Patient Identifier (PID)			
Position	**Element**	**Required?**	**Sample Content**
0	Message Name	Required	PID
1	Set Id	Not Used	
2	Patient Id-External	Not Used	
3	Patient Id-Internal	Required	52068866
4	Alternate Patient Id	Not Used	
5	Patient Name	Required	LASTNAME^FIRSTNAME
6	Mother Maiden Name	Not Used	
7	Date of Birth	Required	19760821
8	Sex	Required	F
9	Patient Alias	Not Used	
10	Ethnic Group	Required	Caucasian
11	Patient Address	Not Used	
12	County Code	Not Used	
13	Phone Number-Home	Not Used	
14	Phone Number-Business	Not Used	
15	Language	Not Used	
16	Marital Status	Not Used	
17	Religion	Not Used	
18	Patient Account Number	Optional	496677136
19	Social Security Number	Not Used	
20	Driver's License	Not Used	
>=21		Not Used	

Patient Visit (PV1)

Patient Visit (PV1)			
Position	**Element**	**Required?**	**Sample Content**
0	Message Name	Required	PV1
1	Set Id	Not Used	
2	Patient Class	Not Used	
3	Assigned Patient Location	Not Used	
4	Admission Type	Not Used	
5	Pre-Admit Number	Not Used	
6	Prior Patient Location	Not Used	
7	Attending Doctor	Optional	FAKE^NAME1
8	Referring Doctor	Optional	FAKE^NAME2
9	Consulting Doctor	Not Used	
10	Hospital Service	Not Used	
11	Temporary Location	Not Used	
12	Pre-Admit Test Indicator	Not Used	
13	Re Admission Indicator	Not Used	
14	Admit Source	Not Used	
15	Ambulatory Status	Not Used	
16	VIP Indicators	Not Used	
17	Admitting Doctor	Not Used	
18	Patient Type	Not Used	
19	Visit Number	Optional	22228615
20	Financial Class	Not Used	
21	Charge Price Indicator	Not Used	
22	Courtesy Code	Not Used	
23	Credit Rating	Not Used	
24	Contract Code	Not Used	
25	Contract Effective Date	Not Used	
26	Contract Amount	Not Used	
27	Contract Period	Not Used	
28	Interest Code	Not Used	
29	Transfer To Bad Debt Code	Not Used	
30	Transfer To Bad Debt Date	Not Used	
31	Bad Debt Agency Code	Not Used	
32	Bad Debt Transfer Amount	Not Used	
33	Bad Debt Recovery Amount	Not Used	

34	Delete Account Indicator	Not Used	
35	Delete Account Date	Not Used	
36	Discharge Disposition	Not Used	
37	Discharge To Location	Not Used	
38	Diet Type	Not Used	
39	Servicing Facility	Not Used	
40	Bed Status	Not Used	
41	Account Status	Not Used	
42	Pending Location	Not Used	
43	Prior Temporary Location	Not Used	
44	Admit Date Time	Not Used	
45	Discharge Date Time	Not Used	
46	Current Patient Balance	Not Used	
47	Total Charges	Not Used	
48	Total Adjustments	Not Used	
49	Total Payments	Not Used	
>=50		Not Used	

Common Order (ORC)

Common Order (ORC)			
Position	**Element**	**Required?**	**Sample Content**
0	Message Name	Required	ORC
1	Order Control	Required	RE
2	Placer Order Number	Optional	12345
3	Filler Order Number	Required	OBR^5_01122019_021633^FVC
4	Placer Group Number	Not Used	
5	Order Status	Not Used	
6	Response Flag	Not Used	
7	Timing Quantity	Not Used	
8	Parent	Not Used	
9	Date Time of Transaction	Not Used	
10	Entered By	Not Used	
11	Verified By	Not Used	
12	Ordering Provider	Optional	Name
13	Enterers Location	Not Used	
14	Callback Phone Number	Not Used	
15	Order Effective Date Time	Not Used	
16	Order Control Code Reason	Not Used	
17	Entering Organization	Not Used	
18	Entering Device	Not Used	
19	Action By	Not Used	
>=20		Not Used	

Common Order (OBR)

Observation Request (OBR)			
Position	**Element**	**Required ?**	**Sample Content**
0	Message Name	Required	OBR
1	Set Id	Required	1
2	Placer Order Number	Optional	73289718
3	Filler Order Number	Required	OBR^5_01122019_021633^FVC
4	Universal Service Id	Required	PFT13^PULMONARY FUNCTION TEST
5	Priority	Not Used	
6	Requested Date Time	Not Used	
7	Observation Date Time	Required	20201105133018
8	Observation End date Time	Not Used	
9	Collection Volume	Not Used	
10	Collection Id	Not Used	
11	Specimen Action Code	Not Used	
12	Danger Code	Not Used	
13	Relevant Clinical Info	Not Used	
14	Specimen Received Date Time	Not Used	
15	Specimen Source	Not Used	
16	Ordering Provider	Optional	NAME
17	Order Callback Phone number	Not Used	
18	Placers Field 1	Not Used	
19	Placers Field 2	Not Used	
20	Filler Field 1	Not Used	
21	Filler Field 2	Not Used	
22	Results Report Status Change dt	Not Used	
23	Charge To Practice	Not Used	
24	Diagnostic Service Sect Id	Required	PF
25	Result Status	Required	F
26	Linked Results	Not Used	
27	Quantity timing	Not Used	
28	Results Copies To	Not Used	
29	Parent Accession Number	Not Used	
30	Transportation Mode	Not Used	
31	Reason for Study	Not Used	
32	Print Result Interpreter	Not Used	

33	Assist Result Interpreter	Not Used	
34	Technician	Not Used	
35	Transcriptionist	Not Used	
36	Scheduled Date Time	Not Used	
>=37		Not Used	

Observation Result (OBX)

Observation Result (OBX)			
Position	**Element**	**Required?**	**Sample Content**
0	Message Name	Required	OBX
1	Set Id	Required	1
2	Value Type	Required	NM or ST or ED
3	Observation Identifier	Required	19868-9_Best^FVC_Best^LN
4	Observation Sub Id	Not Used	
5	Observation Result	Required	2.53
6	Units	Required	L
7	Reference Range	Not Used	
8	Abnormal Flags	Not Used	
9	Probability	Not Used	
10	Nature Of Abnormal Test	Not Used	
11	Observation Result Status	Required	F
12	Date Last Normal Value	Not Used	
13	User Defined Access Checks	Not Used	
14	Date Time of The Observation	Not Used	
15	Procedure's Id	Not Used	
16	Responsible Observer	Not Used	
17	Observation Method	Not Used	
>=18		Not Used	

For each measured parameter, that was selected by the user to be included in the summary PDF report in the parameter settings screen, the following six OBX segments are included:

1. Best measured value
2. Predicted value
3. Percent of the measured value out of the predicted value
4. Lower Limit of Normal (LLN) for that parameter
5. Upper Limit of Normal (ULN) for that parameter
6. Z-score for that parameter (the number of standard deviations between the measured value and the predicted value)

```
OBX|1|ST|19868-9_Best^FVC_Best^LN||5.03|L|||||F||||||
OBX|2|ST|19868-9_Pred^FVC_Pred^LN||4.43|L|||||F||||||
OBX|3|ST|19868-9_%Pred^FVC_%Pred^LN||113.45|%|||||F||||||
OBX|4|ST|19868-9_LLN^FVC_LLN^LN||3.42|L|||||F||||||
OBX|5|ST|19868-9_ULN^FVC_ULN^LN||5.46|L|||||F||||||
OBX|6|ST|19868-9_ZSCORE^FVC_ZSCORE^LN||0.96|None|||||F||||||
```

For tests that support both Pre and Post Bronchodilator measurements, there will be 9 OBX segment per parameter:

1. Best measured value Pre-Bronchodilator
2. Predicted Result
3. Percent of the Pre-Bronchodilator measured value out of the predicted value
4. Lower Limit of Normal (LLN) for that parameter
5. Upper Limit of Normal (ULN) for that parameter
6. Z-score for that Pre-Bronchodilator measured value (the number of standard deviations between the measured value and the predicted value)
7. Best measured value Post-Bronchodilator
8. Percent of the Post-Bronchodilator measured value out of the predicted value
9. Z-score for that Post-Bronchodilator measured value (the number of standard deviations between the Post-Bronchodilator measured value and the predicted value)

```
OBX|1|ST|19868-9_Best_Pre^FVC_Best_Pre^LN||5.03|L|||||F||||||
OBX|2|ST|19868-9_Pred^FVC_Pred^LN||4.43|L|||||F||||||
OBX|3|ST|19868-9_%Pred_Pre^FVC_%Pred_Pre^LN||113.45|%|||||F||||||
```

OBX|4|ST|19868-9_LLN^FVC_LLN^LN||3.42|L|||||F||||||
OBX|5|ST|19868-9_ULN^FVC_ULN^LN||5.46|L|||||F||||||
OBX|6|ST|19868-9_ZSCORE_Pre^FVC_ZSCORE_Pre^LN||0.96|None|||||F||||||
OBX|7|ST|19868-9_Best_Post^FVC_Best_Post^LN||5.03|L|||||F||||||
OBX|8|ST|19868-9_%Pred_Post^FVC_%Pred_Post^LN||113.45|%|||||F||||||
OBX|9|ST|19868-9_ZSCORE_Post^FVC_ZSCORE_Post^LN||0.96|None|||||F||||||

The format of the OBR segment for the encapsulated visit summary pdf report:

OBX|283|ED|58477-1^Pulmonary Function Test Report^VSPULR||^AP^PDF^Base64^…………

The maximum length of the pre encoded data in one encapsulated pdf OBX segment was fixed at 30000 bytes for all PulmOne MiniBox versions prior to version 3.1.3. The encapsulated summary PDF report is usually longer than 30000 bytes, so it was split into multiple OBX segments, and each segment was individually Base64 encoded. The receiving application had to Base64 decode the Base64 encoded content of each encapsulated pdf OBR segment before appending them to recreate the PDF report.
Starting from version 3.1.3 of the PulmOne Minibox, the maximum number of bytes in each OBR segment can be increased using a property in the MiniBox config file, reducing the number of OBR segments required to transmit the encapsulated pdf report.

Sample ORUR01 sent from the MiniBox to the EMR HL7 results interface

```
MSH|^~\&|MiniBox|PulmOne|||202110311513|0|ORU^R01|19_24102021_19
0349||252|||||||||LRI_NG_RN_Profile^HL7^2.16.840.1.113883.9.20^I
SO

PID|||1470||LastName^FirstName||19760827|Male||Caucasian|||||||
||

PV1||||||||||||||||||||||||||||||||||||||||||||||||

ORC|RE||OBR^19_24102021_190349^FVC|||||||||||||||

OBR|1||OBR^19_24102021_190349^FVC|FVC|||20211024190349||||||||||
|||||||PF|F|||||||||||

OBX|1|ST|19868-9_Best^FVC_Best^LN||5.03|L|||||F||||||
```

```
OBX|2|ST|19868-9_Pred^FVC_Pred^LN||4.43|L|||||F||||||

OBX|3|ST|19868-9_%Pred^FVC_%Pred^LN||113.45|%|||||F||||||

OBX|4|ST|19868-9_LLN^FVC_LLN^LN||3.42|L|||||F||||||

OBX|5|ST|19868-9_ULN^FVC_ULN^LN||5.46|L|||||F||||||

OBX|6|ST|19868-9_ZSCORE^FVC_ZSCORE^LN||0.96|None|||||F||||||

OBX|7|ST|20150-9_Best^FEV1_Best^LN||3.92|L|||||F||||||

OBX|8|ST|20150-9_Pred^FEV1_Pred^LN||3.48|L|||||F||||||

OBX|9|ST|20150-9_%Pred^FEV1_%Pred^LN||112.77|%|||||F||||||

OBX|10|ST|20150-9_LLN^FEV1_LLN^LN||2.66|L|||||F||||||

OBX|11|ST|20150-9_ULN^FEV1_ULN^LN||4.25|L|||||F||||||

OBX|12|ST|20150-9_ZSCORE^FEV1_ZSCORE^LN||0.94|None|||||F||||||

OBX|13|ST|19926-5_Best^FEV1/FVC_Best^LN||77.99|%|||||F||||||

OBX|14|ST|19926-5_Pred^FEV1/FVC_Pred^LN||78.58|%|||||F||||||

OBX|15|ST|19926-5_%Pred^FEV1/FVC_%Pred^LN||99.25|%|||||F||||||

OBX|16|ST|19926-5_LLN^FEV1/FVC_LLN^LN||67.04|%|||||F||||||

OBX|17|ST|19926-5_ULN^FEV1/FVC_ULN^LN||88.56|%|||||F||||||

OBX|18|ST|19926-5_ZSCORE^FEV1/FVC_ZSCORE^LN||-
0.09|None|||||F||||||
```

…

Form **W-9**
(Rev. October 2018)
Department of the Treasury
Internal Revenue Service

Request for Taxpayer Identification Number and Certification

▶ Go to *www.irs.gov/FormW9* for instructions and the latest information.

Give Form to the requester. Do not send to the IRS.

Print or type. See **Specific Instructions** on page 3.

1 Name (as shown on your income tax return). Name is required on this line; do not leave this line blank.

PulmOne USA Inc.

2 Business name/disregarded entity name, if different from above

3 Check appropriate box for federal tax classification of the person whose name is entered on line 1. Check only **one** of the following seven boxes.

☐ Individual/sole proprietor or single-member LLC ☑ C Corporation ☐ S Corporation ☐ Partnership ☐ Trust/estate

☐ Limited liability company. Enter the tax classification (C=C corporation, S=S corporation, P=Partnership) ▶ ____

Note: Check the appropriate box in the line above for the tax classification of the single-member owner. Do not check LLC if the LLC is classified as a single-member LLC that is disregarded from the owner unless the owner of the LLC is another LLC that is **not** disregarded from the owner for U.S. federal tax purposes. Otherwise, a single-member LLC that is disregarded from the owner should check the appropriate box for the tax classification of its owner.

☐ Other (see instructions) ▶

4 Exemptions (codes apply only to certain entities, not individuals; see instructions on page 3):

Exempt payee code (if any) ____

Exemption from FATCA reporting code (if any) ____

(Applies to accounts maintained outside the U.S.)

5 Address (number, street, and apt. or suite no.) See instructions.

444 Madison Avenue

Requester's name and address (optional)

6 City, state, and ZIP code

NYC, NY, 10022

7 List account number(s) here (optional)

Part I Taxpayer Identification Number (TIN)

Enter your TIN in the appropriate box. The TIN provided must match the name given on line 1 to avoid backup withholding. For individuals, this is generally your social security number (SSN). However, for a resident alien, sole proprietor, or disregarded entity, see the instructions for Part I, later. For other entities, it is your employer identification number (EIN). If you do not have a number, see *How to get a TIN,* later.

Note: If the account is in more than one name, see the instructions for line 1. Also see *What Name and Number To Give the Requester* for guidelines on whose number to enter.

Social security number

☐☐☐ – ☐☐ – ☐☐☐☐

or

Employer identification number

4 7 – 4 1 6 5 9 4 5

Part II Certification

Under penalties of perjury, I certify that:

1. The number shown on this form is my correct taxpayer identification number (or I am waiting for a number to be issued to me); and
2. I am not subject to backup withholding because: (a) I am exempt from backup withholding, or (b) I have not been notified by the Internal Revenue Service (IRS) that I am subject to backup withholding as a result of a failure to report all interest or dividends, or (c) the IRS has notified me that I am no longer subject to backup withholding; and
3. I am a U.S. citizen or other U.S. person (defined below); and
4. The FATCA code(s) entered on this form (if any) indicating that I am exempt from FATCA reporting is correct.

Certification instructions. You must cross out item 2 above if you have been notified by the IRS that you are currently subject to backup withholding because you have failed to report all interest and dividends on your tax return. For real estate transactions, item 2 does not apply. For mortgage interest paid, acquisition or abandonment of secured property, cancellation of debt, contributions to an individual retirement arrangement (IRA), and generally, payments other than interest and dividends, you are not required to sign the certification, but you must provide your correct TIN. See the instructions for Part II, later.

Sign Here **Signature of U.S. person ▶** **Date ▶** 12/12/20

General Instructions

Section references are to the Internal Revenue Code unless otherwise noted.

Future developments. For the latest information about developments related to Form W-9 and its instructions, such as legislation enacted after they were published, go to *www.irs.gov/FormW9.*

Purpose of Form

An individual or entity (Form W-9 requester) who is required to file an information return with the IRS must obtain your correct taxpayer identification number (TIN) which may be your social security number (SSN), individual taxpayer identification number (ITIN), adoption taxpayer identification number (ATIN), or employer identification number (EIN), to report on an information return the amount paid to you, or other amount reportable on an information return. Examples of information returns include, but are not limited to, the following.

• Form 1099-INT (interest earned or paid)

• Form 1099-DIV (dividends, including those from stocks or mutual funds)

• Form 1099-MISC (various types of income, prizes, awards, or gross proceeds)

• Form 1099-B (stock or mutual fund sales and certain other transactions by brokers)

• Form 1099-S (proceeds from real estate transactions)

• Form 1099-K (merchant card and third party network transactions)

• Form 1098 (home mortgage interest), 1098-E (student loan interest), 1098-T (tuition)

• Form 1099-C (canceled debt)

• Form 1099-A (acquisition or abandonment of secured property)

Use Form W-9 only if you are a U.S. person (including a resident alien), to provide your correct TIN.

If you do not return Form W-9 to the requester with a TIN, you might be subject to backup withholding. See What is backup withholding, *later.*